

ISMAILIA, Egypt

WMO: 624400

Lat: 30.592N

Lon: 32.247E

Elev: 13

StdP: 101.17

Time Zone: 2.00 (E02)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.8	7.9	-0.4	3.7	16.8	1.1	4.1	16.4	12.6	14.9	10.9	16.1	1.3	270	0.406

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.5	37.7	22.1	36.5	22.2	35.3	22.2	25.5	31.1	24.9	30.7	24.4	30.1	5.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.0	27.6	23.4	18.2	27.1	22.9	17.7	26.8	78.5	31.2	75.9	30.5	73.8	30.1	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.1	8.7	7.9	DB	4.8	41.4	1.4	1.6	3.8	42.6	3.0	43.5	2.3	44.4	1.3	45.6	
			WB	3.3	27.0	1.2	1.2	2.5	27.9	1.8	28.5	1.1	29.2	0.3	30.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.1	14.1	15.1	17.5	21.0	24.5	27.3	29.2	29.4	27.4	24.2	19.7	15.6
	DBStd	5.90	2.01	2.58	3.20	3.30	2.52	2.04	1.41	1.28	1.68	2.33	2.42	2.01
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	391	132	96	56	7	0	0	0	0	0	0	12	87
	CDD10.0	4426	128	143	231	329	451	519	597	603	522	439	290	174
	CDD18.3	1774	1	6	28	86	192	269	338	344	272	181	52	3
	CDH23.3	18788	13	64	267	817	1843	2944	4142	4174	2794	1399	304	26
Wind	WSAvg	3.6	3.1	3.7	4.0	4.1	4.0	3.9	3.8	3.7	3.5	3.2	2.8	3.0
	PrecAvg	23	6	4	4	2	1	0	0	0	0	1	1	3
	PrecMax	45	21	12	20	22	4	0	0	0	0	4	16	12
	PrecMin	3	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	12	7	4	5	5	1	0	0	0	0	1	3	4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.1	29.2	33.6	37.5	38.2	39.2	39.0	38.1	38.5	36.0	31.0	26.2
		MCWB	15.1	16.0	17.4	19.1	20.1	22.0	22.8	23.3	21.6	20.5	18.9	15.5
	2%	DB	22.1	24.9	29.1	33.8	35.1	36.9	37.1	37.0	35.4	32.2	28.1	23.3
		MCWB	13.8	14.0	16.2	17.9	19.0	21.6	22.9	23.4	22.2	21.0	18.6	15.0
	5%	DB	20.2	22.2	26.1	30.6	33.1	35.0	35.9	35.9	33.8	30.6	26.4	21.9
		MCWB	13.4	13.5	15.8	16.9	18.9	21.3	22.8	23.2	22.2	20.9	18.0	14.7
	10%	DB	19.0	20.4	23.5	27.9	31.1	33.5	34.9	34.6	32.3	29.2	24.9	20.7
		MCWB	12.8	13.1	14.9	16.4	18.8	21.2	22.9	23.2	22.1	20.6	17.6	14.2
	0.4%	WB	16.6	17.4	19.7	20.7	21.7	24.6	26.0	26.6	25.2	24.2	21.2	17.6
		MCDB	22.0	25.8	28.1	32.6	31.2	31.6	31.9	31.9	30.3	29.4	26.4	22.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	15.0	15.4	17.6	19.0	20.8	23.6	25.2	25.8	24.3	22.9	20.2	16.5
		MCDB	20.2	21.8	26.3	29.3	29.8	30.9	31.3	31.1	29.8	27.7	25.3	21.5
	5%	WB	14.1	14.5	16.4	18.0	20.2	22.9	24.6	25.2	23.7	22.2	19.4	15.6
		MCDB	19.0	20.1	23.6	27.3	28.5	30.2	30.7	30.8	29.3	27.5	24.2	20.5
	10%	WB	13.4	13.7	15.4	17.2	19.7	22.4	24.1	24.6	23.2	21.5	18.5	14.8
		MCDB	18.1	19.0	21.9	25.2	27.5	29.4	30.1	30.2	28.8	27.0	23.3	19.4
Mean Daily Temperature Range	5% DB	MDBR	10.3	10.4	11.2	12.5	13.0	12.7	12.0	11.5	11.2	10.8	11.0	10.5
		MCDBR	12.1	13.4	14.7	16.7	15.8	14.7	13.3	12.6	12.5	12.0	12.1	12.1
		MCWBR	6.0	5.4	5.1	5.2	4.2	3.6	3.3	3.3	3.3	3.6	4.6	5.8
	5% WB	MCDBR	11.2	11.6	13.4	14.3	13.7	13.3	12.3	11.9	11.7	11.4	11.4	11.2
		MCWBR	6.1	5.4	5.2	5.1	4.0	3.6	3.5	3.4	3.4	3.7	4.7	6.0
Clear-Sky Solar Irradiance	taub		0.402	0.441	0.487	0.534	0.536	0.456	0.451	0.454	0.459	0.468	0.410	0.395
	taud		2.084	1.964	1.852	1.764	1.787	2.027	2.057	2.052	2.029	1.998	2.135	2.131
	Ebn at Noon		801	804	796	776	775	837	839	833	812	768	788	786
	Edh at Noon		134	166	199	225	222	174	168	167	164	158	127	123
All-Sky Solar Radiation	RadAvg		3.45	4.30	5.63	6.77	7.62	8.16	7.98	7.36	6.32	4.97	3.81	3.21
	RadStd		0.11	0.20	0.22	0.24	0.22	0.10	0.09	0.09	0.08	0.13	0.11	0.10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	+0.60	N/A	-1.36	N/A	N/A	N/A	N/A	-50	+212	+171

Nomenclature: See separate page